University of Dhaka

Fourth Year BS (Honours) 2023-24

Assignment 01

Subject: Applied Mathematics

Course No.: AMTH 450 Course Title: Math Lab IV

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NB: You will get all the necessary files here at:
https://github.com/NirabHossain/python-assignment-documents

1. Write a program to calculate grade for a student of the Department of Applied Mathematics, University of Dhaka. Note that the grades are calculated by summing up each student's average of first incourse and second incourse and attendance mark with the final examination marks. You are to take input of some random first incourse, second incourse, attendance and final marks from file and show the grade with the help of the following table.

Marks	Letter Grade	Grade Point
≥ 80	A^+	4.00
≥ 75	A	3.75
≥ 70	A^-	3.50
≥ 65	B^+	3.25
≥ 60	B	3.00
≥ 55	B^-	2.75
≥ 50	C^+	2.50
≥ 45	C	2.25
≥ 40	D	2.00
<40	F	0.00

If the user enters a value out of range, print a suitable error message and exit. For the test, enter a score of 185.
Your output should look like:

Roll	Letter Grade	Grade Point
80	A^+	4.00
81	A	3.75
82	A^-	3.50

2. Read through and parse the file *regex-sum-42.txt* with text and numbers. Extract all the numbers in the file and compute the sum of the numbers.

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3. Open the file *amth.txt* and read it line by line. For each line, split the line into a list of words using the *split*() method. The program should build a list of words. For each word on each line check to see if the word is already in the list and **if not append it to the list**. When the program completes, *sort* and *print* the resulting words in **alphabetical order**.

4. Write a program that prompts for a file name, then opens that file and reads through the file *mbox.txt*, looking for lines of the form:

X-DSPAM-Confidence: 0.8475

Count these lines and extract the floating point values from each of the lines and compute the *average* of those values and produce an output as shown below. Do not use the *sum*() function or a variable named sum in your solution.

5. Open the file *mbox.txt* and read it line by line. When you find a line that starts with *'From'* like the following line:

From stephen.marquard@uct.ac.za Sat Jan 5 09:14:16 2008

You will *parse* the From line using *split*() and print out the second word in the line (i.e. the entire address of the person who sent the message). Then print out a count at the end. If you test your code with the file *mbox-short.text*, your output should look like:

stephen.marquard@uct.ac.za
louis@media.berkeley.edu
zqian@umich.edu
rjlowe@iupui.edu
There were 27 lines in the file with From as the first word

6. Write a program to read through the *mbox.txt* and figure out who has the sent the greatest number of mail messages. The program looks for *'From'* lines and takes the second word of those lines as the person who sent the mail. The program creates a **Python dictionary** that maps the sender's mail address to a count of the number of times they appear in the file. After the dictionary is produced, the program reads through the *dictionary* using a *maximum loop* to find the most prolific committer. If you test your code with the file *mbox-short.text*, your output should look like:

cwen@iupui.edu 5